Canadian Journal of Cardiology 30 (2014) 544–552

Review

Systemic Implementation Strategies to Improve Hypertension: The Kaiser Permanente Southern California Experience

John J. Sim, MD,[a] Joel Handler, MD,[b] Steven J. Jacobsen, MD, PhD,[c] and Michael H. Kanter, MD[b]

[a] Division of Nephrology and Hypertension, Kaiser Permanente Los Angeles Medical Center, Los Angeles, California, USA

[b] Quality and Clinical Analysis, Southern California Permanente Medical Group, Kaiser Permanente Southern California, Pasadena, California, USA

[c] Department of Research and Evaluation, Kaiser Permanente Southern California, Pasadena, California, USA

ABSTRACT

The past decade has seen hypertension improving in the United States where control is approximately 50%. Kaiser Permanente has mirrored and exceeded these national advances in control. Integrated models of care such as Kaiser Permanente and the Veterans Administration health systems have demonstrated the greatest hypertension outcomes. We detail the story of Kaiser Permanente Southern California (KPSC) to illustrate the success that can be achieved with an integrated health system model that uses implementation, dissemination, and performance feedback approaches to chronic disease care. KPSC, with a large ethnically diverse population of more than 3.6 million, has used a stepwise approach to achieve control rates greater than 85% in those recognized with hypertension. This was accomplished through systemic implementations of specific strategies: (1) capturing hypertensive members into a hypertension registry; (2) standardization of blood pressure measurements; (3) drafting and disseminating an internal treatment algorithm that is evidence-based and is advocating of combination therapy; and (4) a multidisciplinary approach using medical assistants, nurses, and pharmacists as key stakeholders. The infrastructure, support, and involvement across all levels of the health

RÉSUMÉ

Au cours de la dernière décennie, les États-Unis ont connu une amélioration de l'hypertension artérielle où la maîtrise se situe à près de 50 %. La Kaiser Permanente a reflété et dépassé ces avancées nationales en matière de maîtrise. Les modèles de soins intégrés comme la Kaiser Permanente et les systèmes de santé de la Veterans Administration ont mis en évidence les meilleurs résultats sur l'hypertension artérielle. Nous décrivons en détail l'histoire de la Kaiser Permanente Southern California (KPSC) pour illustrer le succès qu'il est possible d'atteindre au moyen d'un modèle intégré des systèmes de santé qui utilise des approches de rétroaction sur la mise en place, la diffusion et la performance des soins en maladies chroniques. La KPSC qui compte une vaste population d'ethnies diverses de plus de 3,6 millions d'individus a utilisé une approche par étape pour atteindre des taux de maîtrise au-dessus de 85 % chez ceux qui sont connus pour l'hypertension artérielle. Cela a été réalisé par la mise en place de stratégies particulières : 1) la saisie au registre de l'hypertension artérielle des membres hypertendus; 2) la standardisation des mesures de la pression artérielle; 3) l'élaboration et la diffusion d'un algorithme thérapeutique interne qui est fondé sur les preuves et qui préconise la

Hypertension affects 1 billion people worldwide accounting for approximately 25%-30% of the adult population. It is the most common chronic condition in the Western hemisphere and the leading reason for ambulatory medical care visits.[1-3] Hypertension is also the most common treatable condition in that controlling it can modify and ameliorate risks for vascular disease outcomes and mortality.[4-9] In Canada and the United States, hypertension prevalence is greater in those with cardiovascular disease risk.[10] In the United States, the awareness, treatment, and subsequent control of hypertension has been historically poor.[1,11-13] The overall control rate among the hypertension population floundered between 20% to 30% in the decades of the 1980s and 1990s.[11] By current estimates in which 68 million people have hypertension,[13] this translates to hundreds of thousands of preventable deaths every year.[14] Moreover, there is a financial burden in the billions of dollars annually that is attributed to hypertension and its untoward consequences.[15]

The reasons for poor control of hypertension are multifaceted. The sources behind these reasons lie within the health care and social environment, the health care providers, and the patients themselves.[16-18] On the population level, screening for and identifying hypertension is challenging because of variability in access to health care and the utilization of

Received for publication October 10, 2013. Accepted January 5, 2014.

Corresponding author: Dr John J. Sim, Division of Nephrology and Hypertension, Kaiser Permanente Los Angeles Medical Center, 4700 Sunset Blvd, Los Angeles, California 90027, USA. Tel.: +1-323-783-4368; fax: +1-323-783-8288.

E-mail: John.j.sim@kp.org

See page 551 for disclosure information.

0828-282X/$ - see front matter © 2014 Canadian Cardiovascular Society. Published by Elsevier Inc. All rights reserved.
http://dx.doi.org/10.1016/j.cjca.2014.01.003

system with rapid and continuous performance feedback have been pivotal in ensuring the follow-through and maintenance of these strategies. The KPSC hypertension program is continually evolving in these areas. With these high control rates and established infrastructure, they are positioned to take on different innovations and study models. Such potential projects are drafting strategies on resistant hypertension or addressing the concerns about overtreatment of hypertension.

multithérapie; 4) une approche multidisciplinaire utilisant les assistants médicaux, les infirmiers et les pharmaciens comme principaux intervenants. L'infrastructure, le soutien et la participation à tous les niveaux du système de santé, et la rétroaction rapide et continue de la performance ont été essentiels pour assurer le suivi et le maintien de ces stratégies. Le programme d'hypertension artérielle de la KPSC est en évolution constante dans ces domaines. Par ces taux de maîtrise élevés et cette infrastructure établie, ils sont en mesure d'assumer les différents modèles d'études et innovations. Ces projets potentiels élaborent des stratégies sur l'hypertension artérielle réfractaire ou répondent aux questions liées au surtraitement de l'hypertension.

resources to effectively capture and manage this population.[17] Canada and the United States face these challenges whether it is in the form of access to physicians or the timeliness of care.[18,19] Among physicians, there is often inertia (unwillingness or indifference) about initiating treatment or escalating therapy to obtain blood pressure control.[20] There are differing views on the appropriate degree of blood pressure control and on the treatment strategies which has led to heterogeneity in practice patterns. Last, individual patients contribute to poor control because of reasons such as nonadherence to lifestyle, medications, and follow-up with their physicians.[21,22] Health literacy is also an important contributor to poor health outcomes that is often overlooked in chronic disease management.[23]

The encouraging news is that in the past decade there have been steady improvements in hypertension awareness and control in the United States. Recent estimates from 2008 suggest that 50% of all hypertensive individuals are controlled and 62% of those treated are controlled.[1] Kaiser Permanente has mirrored the national trend in improvement of hypertension control and has surpassed the nation in the absolute rates of awareness, treatment, and control (Fig. 1). As of 2012, 85% of identified hypertensive individuals within Kaiser Permanente have controlled blood pressure.[24] Across all ages, races, and sexes, hypertension control has exceeded 80%. The progress in hypertension control over the years has coincided with a marked improvement in adverse cardiovascular outcomes among Kaiser Permanente members.[25] How did it happen? What changes were implemented? The answer lies in a concerted effort with the support and involvement of administration, the clinical workforce, and operational leadership that led to a program of effective implementation, dissemination, and continuous performance feedback. Our story follows.

Kaiser Permanente Southern California

Kaiser Permanente is an integrated health system comprised of the health plan, hospitals, and physician group. Founded in 1945, it is comprised of 8 geographical regions across the continental United States and Hawaii. Each region operates independently but also interdependently in terms of collaborating on Kaiser Permanente national goals and guidelines. As a prepaid integrated health plan, members have similar access to health care in terms of office visits, medications, and medical supplies. The internal network of referrals also ensures similar access to and levels of subspecialty care. The current membership of Kaiser Permanente exceeds 8 million individuals, with the largest proportion derived from the California regions.

Kaiser Permanente Southern California (KPSC) is a Kaiser Permanente region established in 1953. It is comprised of 14 medical centres and more than 200 satellite medical office buildings. Geographically, the region spans from Bakersfield to San Diego. As of August, 2013, the health system exceeded 3.6 million members. Complete health care encounters are tracked using a common electronic health record (EHR) system. This includes pharmacy information because more than 95% of members are able to obtain their medications from KPSC pharmacies. All laboratory data, diagnostic and procedure codes, and vital sign assessments, including blood pressure measurements and body mass index are collected in our EHR as part of routine clinical care encounters.

The KPSC population is ethnically and socioeconomically diverse, reflecting the general population of the catchment area and the state of California.[26] Approximately 78% of KPSC members have graduated high school or have received higher level education beyond high school. In terms of economic status, 80% have income levels above the poverty line.[26] Thus, it is likely representative in terms of the different racial/ethnic makeup of the United States and elsewhere (Fig. 2). Among the 2.4 million adults, hypertension is prevalent in approximately 28% of the population (Handler et al., unpublished data)[27] which is comparable with the 29% estimated in National Health and Nutrition Examination Survey (NHANES).[1]

History of KPSC Hypertension

Up until 2000, KPSC did not have a hypertension program per se. The rate of control was similar to the estimated national average and below the average reported in the National Committee for Quality Assurance Healthcare Effectiveness Data and Information Set. The state of hypertension care at the time provided motivation to improve. In addition to a goal of exceeding the average Healthcare Effectiveness Data and Information Set measures, KPSC has sought to become the national leader in health care delivery. Hypertension control became one of the cornerstones of the clinical strategic goals. The successful management of chronic conditions such as hypertension would translate into the prevention of many adverse events and persons lives saved.

Fortunately, the infrastructure for a systematic implementation to improve hypertension control was already in place. There was an integrated health system model with detailed capture of clinic care and follow-up. The organizational focus of KPSC was to tackle chronic diseases and conditions using a large population care-focused model, with systems-based

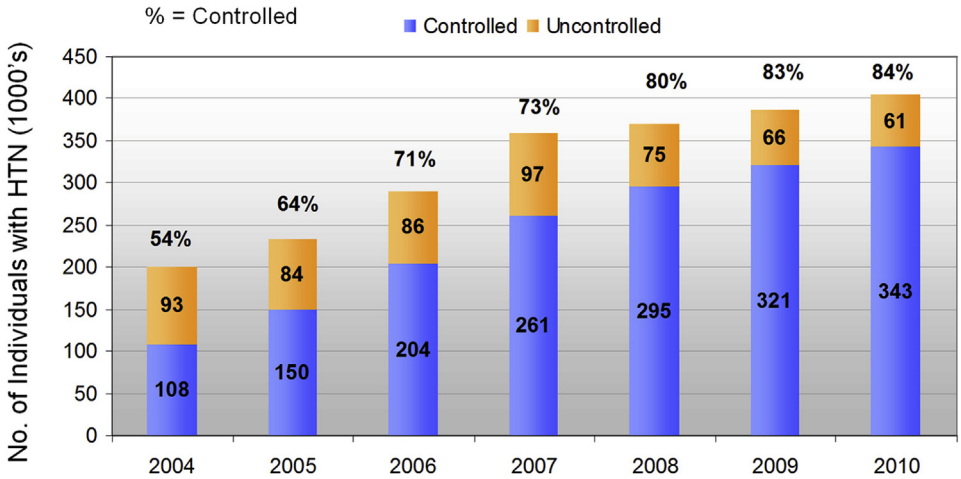

Figure 1. Kaiser Permanente Southern California hypertension control rates. HTN, hypertension.

interventions. Around 2004 KPSC reorganized its focus on managing individual chronic diseases to a system focused on creating reliable processes of care for multiple chronic diseases simultaneously. This system was called Complete Care. It took advantage of the fact that most chronic diseases have similar elements of care that are promptly required.[28]

KPSC now had an organized comprehensive system that was fast to derive information and act on it. Thus, KPSC was well positioned to implement new strategies and rapidly disseminate them. Monthly performance reports promoted the recognition and dissemination of best practices, and a reduction in the variability of performance. Most importantly, they had the collaborative support and buy-in of people at all levels of the health system including administrators, operational leaders, and the clinical workforce.

The Hypertension Program and Implementation

The story of the KPSC hypertension program is an evolving one in which a step-wise approach over the past decade has led to achieving unprecedented hypertension control rates. It started simply with asking the "who, what, and how." This led to transformative steps directed at the system level which included: (1) creation of a hypertension registry; (2) standardization of blood pressure measurements; (3) creation of an internal treatment algorithm; and (4) the

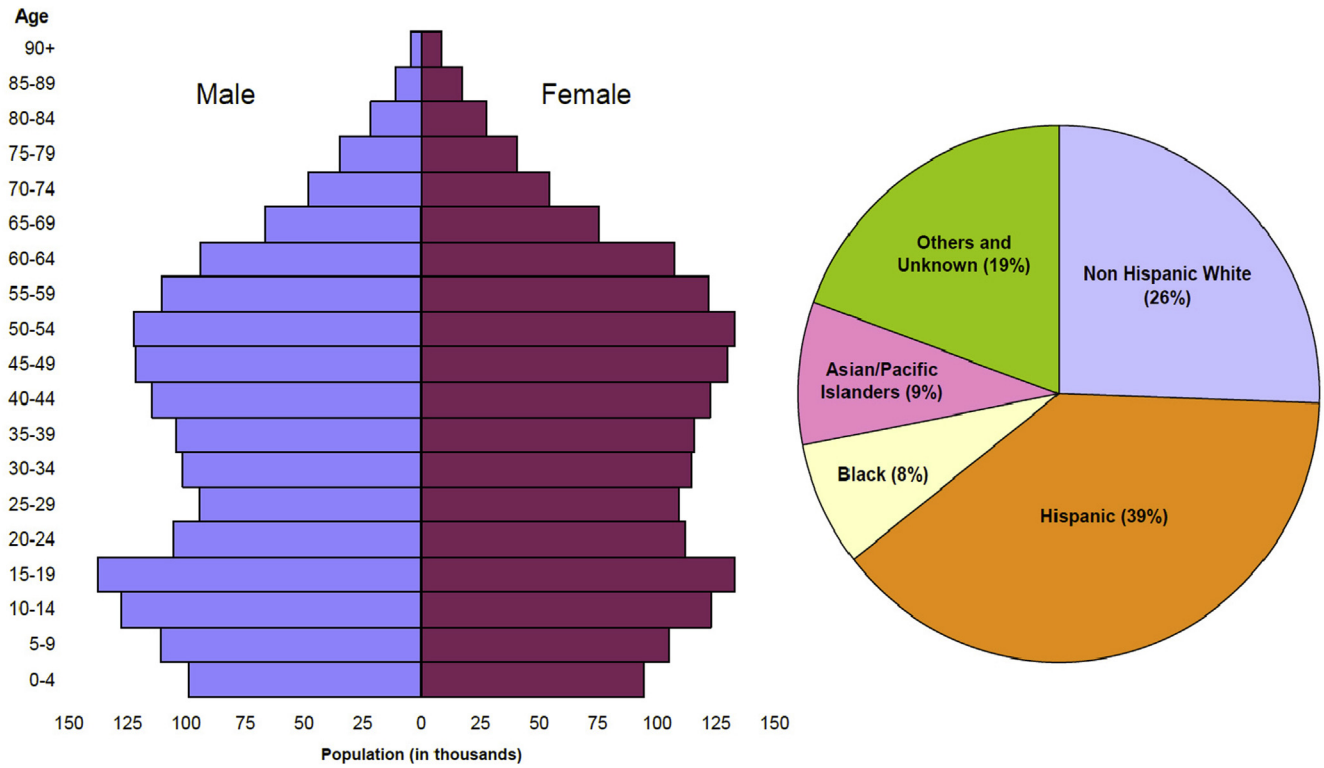

Figure 2. Kaiser Permanente Southern California population overview.

embracement of a multidisciplinary approach of stakeholders, including medical assistants, nurses, and pharmacists. In addition, steps were directed specifically at physicians and patients to motivate them and involve them in all aspects of the implementation steps. What ensued was a synergized renovation of the hypertension program.

Systems-based interventions

Hypertension registry. The first step to improve hypertension was to identify the "who." This meant finding all the individuals who had hypertension at KPSC and ensuring that they were followed and treated. The creation of the hypertension registry was the means to that end. The first hypertension registry was created by Kaiser Permanente Northern California in the year 2000. KPSC followed with their registry, created in 2004. The goal of the registry was to accurately and reliably capture all the hypertensive individuals into 1 database where continual access to their information would be available, individually and in aggregate. The hypertension registry would provide information on the prevalence and description of those with hypertension to better enable the drafting and implementation of strategies to improve. Inclusion in the registry was based on either of 2 criteria: (1) 2 separate International Classification of Diseases, 9th Revision codes for hypertension within a 365-day period; or (2) 1 hypertension code plus at least 1 of the following: a prescription for an antihypertensive medication or a diagnosis of stroke, chronic kidney disease, coronary artery disease, or diabetes mellitus. Thus, the importance of the clinician to recognize and document hypertension became more prominent.

The creation of the hypertension registry coincided with an improvement in recognition of hypertensive individuals. The size of the KPSC hypertension registry had grown from 400,000 in early 2003 to more than 670,000 by the year 2012 (Handler et al., unpublished data).[27] Even the length of time to recognize and code hypertension decreased after the registry was created.[29] In terms of outcomes, a 65% growth in the registry corresponded with a 30% increase in hypertension control.

Standardization of blood pressure measurements. Hypertension was the condition but the validity of blood pressure measurement was the "what" that needed to be addressed first. Quality control was an issue. Variations in blood pressure measurement techniques had to be minimized and blood pressure information had to be reliable. More than 2.3 million blood pressure measurements are performed monthly by nurses and medical assistants across the various health care encounters within more than 200 medical facilities.[30] A model to standardize blood pressure assessment to ensure accuracy and reliability was needed.

KPSC drafted a blood pressure competency model using the following 4 metrics: (1) bare arm during measurement; (2) arm supported at heart level; (3) use of an appropriate cuff size; and (4) no talking during the measurement. All clinical staff were not only trained on the blood pressure competency model but they were and still are continuously evaluated on them. The evaluations are performed through a validated peer review and performance feedback auditing model.[31] Approximately 5 peer reviews occur weekly in each office building using the 4 metrics described. The nurses and medical assistants get immediate feedback and coaching from their department administrators based on the assessments. This peer performance model led to a reduction in the 4 metrics technique errors by 40%.[30]

Consistent with ensuring accuracy, KPSC also sought to ensure the accuracy of the hypertension diagnoses by emphasizing the need to repeat blood pressure measurements in those with initially high numbers. Handler et al., reported that in the NHANES, approximately 20% of initially high blood pressures are reclassified with subsequent repeat measurements.[32] Medical assistants receive best practice alerts to repeat the blood pressure when the initial blood pressure exceeds normal. The use of performance feedback reports to department administrators has improved second blood pressure measurements by medical assistants to 94% when the first blood pressure is elevated (Handler et al., unpublished data). The reasoning is that physicians are more likely to acknowledge and act on what they view to be competent blood pressure elevations.

The treatment algorithm

To help ensure homogeneity of practice delivered, the hypertension treatment had to be standardized as well. This meant that an internal treatment guideline was needed. A medication treatment algorithm that was simple, easy to follow, and evidence-based would be ideal in that it would have more clinician buy-in and also improve patient adherence. Thus, with input from Kaiser Permanente experts and the reports of the Joint National Committee on Prevention, Detection, Evaluation, and Treatment of High Blood Pressure, a Kaiser Permanente Hypertension treatment algorithm was created by the Care Management Institute and advocated to all providers. The first algorithm in 2001 supported a thiazide diuretic as the first line of therapy. Since then it has undergone 4 modifications and the most current algorithm advocates combination therapy with an angiotensin-converting enzyme inhibitor and a thiazide diuretic as first-line therapy, regardless of the stage of hypertension (Fig. 3).[24]

The treatment algorithm was readily available to all physicians. It was provided to physicians in the form of plastic reference cards and posters within offices and exam rooms. Physicians were further educated on the algorithm through continuing medical education (CME) activities such as the KPSC regional hypertension symposium and departmental educational activities such as grand rounds and journal clubs. For example, a CME-accredited debate was presented to primary care physicians in which the "merits" for using β-blockers as first-line therapy were debated. This coincided with the changes in the algorithm in which β-blockers were removed as first-line therapy.

KPSC firmly advocates combination and combination pill therapy. One of the most impactful aspects of the algorithm is the support for combination medication as the first-line therapy regardless of the stage of hypertension. Feldman et al., in the **S**implified **T**reatment **I**ntervention **t**o **C**ontrol **H**ypertension (STITCH) study, soundly demonstrated that initiation of a 2-drug combination regimen led to faster and improved control of hypertension.[33] The superiority of combination therapy was further validated in a study that examined the experience from Kaiser Permanente.[34] At KPSC, the trend for improvement in

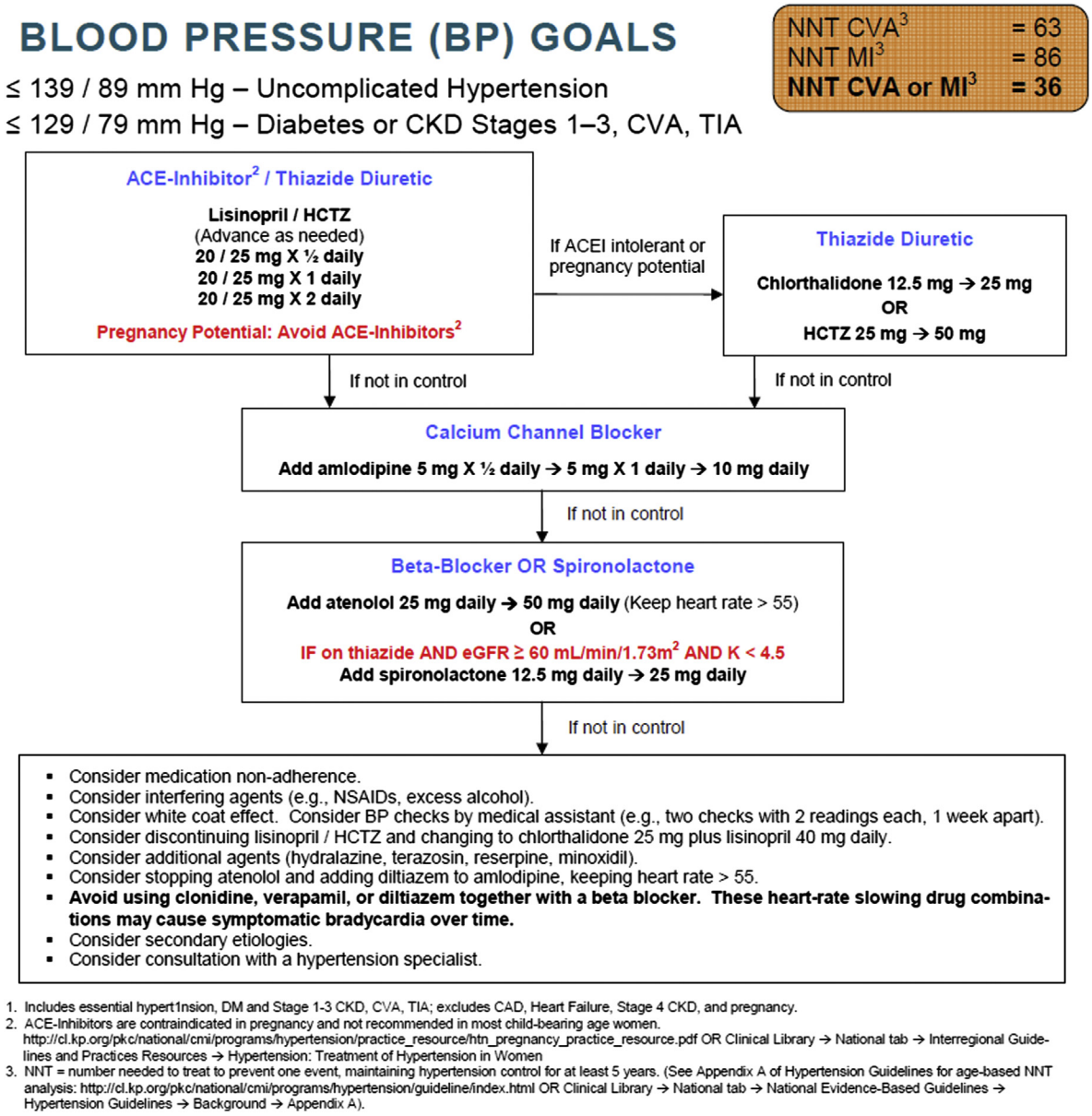

Figure 3. Kaiser Permanente hypertension algorithm. ACE, angiotensin-converting enzyme; ACEI, angiotensin-converting enzyme inhibitor; BP, blood pressure; CKD, chronic kidney disease; CVA, cerebrovascular accident; eGFR, estimated glomerular filtration rate; HCTZ, hydrochlorothiazide; MI, myocardial infarction; NNT, number needed to treat; NSAID, nonsteroidal anti-inflammatory drug; TIA, transient ischemic attack.

control essentially mirrored the increase in use of the lisinopril/hydrochlorothiazide combination pill (Fig. 4). The initial recommendation of lisinopril/hydrochlorothiazide pill in the 2005 version of the algorithm was associated with a 15% increase in hypertension control the following year. Additionally, the 2-drug initial regimen does not appear to add an extra medication burden because the average hypertensive individual usually requires 2 or more medicines to control their blood pressure.[35,36] The combination of these agents into a single pill also translated into fewer copayments and improved medication adherence compared with use of 2 separate medications.[37]

The effect of nonphysician providers

The "how" entailed a comprehensive multidisciplinary approach by many stakeholders. The easy access to nonphysician providers to help manage hypertension has been a cornerstone of the coordinated effort for hypertension control. Members are encouraged and solicited to come into the office for a blood pressure measurement with a medical assistant, with no copay required. The medical assistant visits accomplish several things including blood pressure monitoring, review of antihypertensive medications, and identification of those who need intervention. They can alert nurse practitioners, pharmacists, and physicians about the uncontrolled blood pressures.

The nurses and pharmacists, under the guidance and advice of the physicians and panel managers can directly implement medication titrations for the patients. In addition, they have a means to follow up on the interventions through the medical assistant visits. The use of an automated

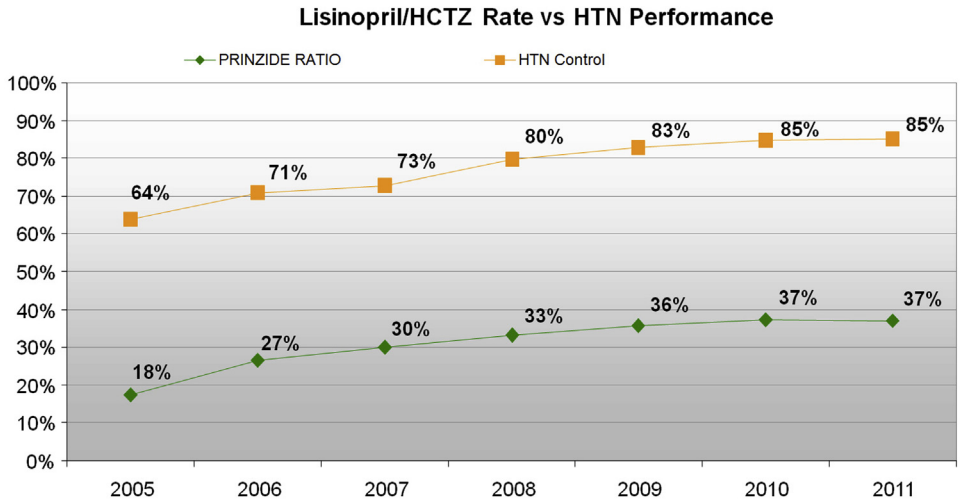

Figure 4. Combination pill use and hypertension control at Kaiser Permanente Southern California. Since 2005, when the combination of lisinopril/HCTZ was advocated, hypertension control rates have steadily increased, paralleling the proportion of those prescribed the lisinopril/HCTZ combination pill. HCTZ, hydrochlorothiazide; HTN, hypertension.

telephone message reminder also improved the adherence of patients to any medication changes.[38] Overall, the hard work and perseverance of this multidisciplinary effort was likely the most impactful step in making a difference on improving patient outcomes. This is the point of care where most of the 'heavy lifting' in the KPSC hypertension program implementations occurred.

Physician-based interventions

Specific implementation strategies directed toward physicians also helped buy-in and participation in the hypertension initiatives. There was extensive feedback directed at overcoming physician inertia and practicing to the standards of the organization. Physicians were encouraged to attend educational conferences and activities on hypertension. Educational time and CME credits were provided by the organization for the physicians to attend. Across different specialties, KPSC regionally advocated for those managing hypertension to obtain certification as a hypertension specialist through the American Society of Hypertension. During the period of 2005-2010, small financial incentives were also offered to physicians to obtain a prespecified level of hypertension control in their panel. Individual performance feedback was given to physicians on a regional, office, and individual level on a quarterly basis. Lower-performing physicians were given additional education, mentoring, and coaching as appropriate.

All physicians were aided in recognizing hypertension. Comprehensive blood pressure measurements were performed and available at all visitations including nonprimary care subspecialty clinics and urgent care. The effort was part of the KPSC Proactive Office Encounter, which was a systematic approach to preventive and chronic care at every patient encounter.[39] The EHR reinforced this initiative because blood pressures $\geq$ 140/90 resulted in a pop-up reminder to the clinician that the values exceeded normal ranges. These types of encounters led to the clinicians becoming more aware and attentive about hypertension.

Medication adherence or the lack thereof is one of the biggest challenges to hypertension control.[40-44] Up to 50% of hypertensive patients are likely to discontinue therapy within 1 year and only a minority (< 40%) of hypertensive subjects have been shown to continue their medicines long-term.[45-47] An even more difficult challenge is to determine and directly measure adherence. Because the KPSC EHR captures medications prescribed and filled for 95% of the members, more reliable objective measures of medication adherence can be obtained. Each clinician has access to the pharmacy records in every encounter including a calculated medication refill and adherence rate based on date and supply of medicines prescribed. The days of supply remaining are also calculated. Thus, patient nonadherence to medicines is readily detected and can be addressed early as a reason for uncontrolled blood pressure.

Through these efforts, KPSC made it easier for physicians to treat hypertension. Physicians were provided with the support of nurses, medical assistants, and readily available information to make treatment-related decisions for their patients. The EHR helped them with reminders about uncontrolled blood pressure and also with information about patient medication adherence. Any histories of adverse reactions to past medicines were also readily available in the EHR. This type of infrastructure with the EHR and access to patients' records and the patients themselves are very comparable with the system of hypertension treatment across the United States Veterans Administration (VA) health system.[48,49] However, the VA system is very different compared with KPSC in that it is comprised of a homogeneous population predominately of men and very little representation of ethnic minorities.[50]

Patient-based interventions

Patient-oriented strategies were implemented to address barriers experienced by the patients themselves. Initiatives were undertaken to address disparities in care, medication adherence, and health literacy. The black population historically had low hypertension control rates outside and within KPSC. Although white/Caucasian hypertension control rates consistently exceeded 85% in the past several years, rates of

the black population hovered at or below 80% within the same periods.[27,36] A well-recognized contributor to hypertension is the role of salt and volume in this subpopulation.[36,51] Thus, educational interventions with emphasis on a low-salt diet were heavily promulgated in this population. In addition, the providers were educated on the importance of diuretic agents as the mainstay of therapy especially in patients who were diuretic-naive.

Patient-centreed programs have been initiated to address health literacy and to communicate information about hypertension. Education programs are given to patients in the form of one-to-one teaching, peer group sessions, and educational videos.[52] The peer group-focused meetings are particularly helpful because members are usually more open about their knowledge deficits to their peers and learn more effectively teaching each other. They can also address similar cultural barriers of which their physicians might not be aware, and in the same language. The concordance in language between providers and patients has an effect on hypertension outcomes. At KPSC, where Spanish is commonly spoken as the first language, an increase in the proportion of Spanish-speaking patients who had Spanish-speaking doctors was associated with a closing of the disparity in hypertension control for that subpopulation (Handler et al., unpublished data).

Sustaining success

The implementation steps detailed herein have brought KPSC hypertension care to unprecedented levels for the health system. The level of success have mirrored Kaiser Permanente Northern California and the other Kaiser Permanente regions.[24] The work continues in an effort to constantly improve. Performance feedback across all providers, medical offices, and hospitals is constantly given. This information is available to everyone within KPSC to encourage healthy competition. Audits to maintain quality care continue. These efforts strive to maintain the successful hypertension control rates and to push for higher control rates. The sharing of best practices and innovations are always encouraged because the organization has proven that it can rapidly implement and disseminate change.

Integrated Health System Models: Kaiser Permanente and United States VA

The hypertension control rates at Kaiser Permanente might be the highest within the United States. This is notable given the large size of the health system and diverse member population. The KPSC hypertension story also speaks to the integrated model of health care in chronic disease management. Overall, an integrated health care system model appears best positioned to succeed in hypertension management. The fact that the United States VA health system has experienced a similar trend of improvement and sustained success in hypertension control further supports is assertion. Within the past decade, the VA integrated health system has seen their hypertension control increase from 46% in the year 2000 to its current rate of 78%.[53,54]

The VA hypertension management is comparable with the Kaiser Permanente system in many ways. The VA also has a comprehensive EHR that has recorded blood pressure and allowed it be to accessible on a wide scale since the year 1998. This was earlier than Kaiser Permanente which began incorporating vital signs into their EHR in the year 2005. The VA EHR has a mechanism to provide reminders to physicians on blood pressure control. It also gives feedback to patients on scheduling and medication refills.[53] Financial considerations is not a barrier within the VA because members have readily available access to their providers with minimal to no cost. The VA hypertension treatment guideline is Joint National Committee on Prevention, Detection, Evaluation, and Treatment of High Blood Pressure-based, advocating diuretic agents as first-line treatment. They have a proven effective infrastructure using nonphysician providers to assume a significant proportion of hypertension care.[55] Regular performance feedback with incentives for providers is also incorporated into their hypertension management programs.

Overall, the Kaiser Permanente health system is not the same as the VA system as evidenced by the differences in control rates (85% vs 78%). The VA does not have a hypertension registry per se. In addition, the VA treatment guideline advocates monotherapy with diuretic agents as first-line therapy unless they are stage 2 or higher compared with Kaiser Permanente, which advocates combination therapy as first-line treatment for all stages of hypertension. Although the VA does instruct providers on the blood pressure measurement techniques, KPSC performs weekly audits on these techniques, which has led to significant reductions in technique variability.[30] KPSC also promotes competition and identifies best practices within medical centres. This has led to more effective processes being rapidly disseminated across the health system.

Ultimately, the difference in the Kaiser Permanente and VA health systems lies in the member population. The VA population exceeds 22 million and the Kaiser Permanente membership is around 8 million. Although KPSC has a sex-balanced population (Fig. 2), the VA population has less than 10% women. The racial/ethnic makeup is different as well, with fewer than 10% of the VA population comprised of non-black minorities.[54] Thus, the size and characteristics of the VA population alone might account for the differences in hypertension control because the 2 integrated health systems have many similarities.

Future Direction

Looking ahead to the direction of KPSC hypertension control is intriguing because of the possibilities. KPSC is positioned to implement strategies that could integrate the current technologies and the changing health care environment. There will be developments in delivery and efficiency of the current programs. For example, efforts are under way to better integrate the automatic telephone reminders into its model. The use of home blood pressure teletransmission is being explored as a means to improve patient self-management. These build on our findings that e-mail communication with members was associated with better hypertension control rates.[56] These changes will likely come together to become a tool for faster and more efficient care. The path is being established toward a form of medical home type of care delivery model for the hypertension population.

The information and the experience that has been collected from the hypertension registry and the implementation programs will be used to help shape clinical practice moving forward. For instance, approximately 13% of the hypertension population met the criteria for resistant hypertension. Another 8% had uncontrolled blood pressure despite using 3 or more medicines.[27] Although the treatment algorithm and hypertension management strategies have been successful for most, there remains a subpopulation in which different or more individualized care is warranted. KPSC appears well positioned to develop, implement, and study such new clinical care models.

The success in control of such a large proportion of the hypertension population also raises concerns for overtreatment in some. Recent studies have suggested that hypertensive individuals might not benefit from aggressive lowering beyond recommended threshold levels. KPSC data have actually demonstrated worse outcomes in patients whose blood pressure was less than 130 systolic and 60 mm Hg diastolic after treatment (Sim et al., unpublished data). This has led to the development of a "safety net" for the hypertension population through which patients with systolic blood pressures less than 110 mm Hg are identified to down-titrate medication. If successful, such a program would have implications on hypertension outcomes and health delivery.

Acknowledgements

The authors thank Drs William Cushman (Chief of Preventive Medicine, Memphis VA Medical Center) and Csaba Kovesdy (Chief of Nephrology, Memphis VA Medical Center) for their insight and assistance describing the United States VA health system hypertension program.

Disclosures

The authors have no conflicts of interest to disclose.

References

1. Egan BM, Zhao Y, Axon RN. US trends in prevalence, awareness, treatment, and control of hypertension, 1988-2008. JAMA 2010;303:2043-50.

2. Cherry DK, Hing E, Woodwell DA, Rechtsteiner EA. National Ambulatory Medical Care Survey: 2006 summary. Natl Health Stat Report 2008:1-39.

3. Wolf-Maier K, Cooper RS, Banegas JR, et al. Hypertension prevalence and blood pressure levels in 6 European countries, Canada, and the United States. JAMA 2003;289:2363-9.

4. Effects of treatment on morbidity in hypertension. Results in patients with diastolic blood pressures averaging 115 through 129 mm Hg. JAMA 1967;202:1028-34.

5. Klag MJ, Whelton PK, Randall BL, et al. Blood pressure and end-stage renal disease in men. N Engl J Med 1996;334:13-8.

6. Prevention of stroke by antihypertensive drug treatment in older persons with isolated systolic hypertension. Final results of the Systolic Hypertension in the Elderly Program (SHEP). SHEP Cooperative Research Group. JAMA 1991;265:3255-64.

7. Turnbull F, Neal B, Ninomiya T, et al. Effects of different regimens to lower blood pressure on major cardiovascular events in older and younger adults: meta-analysis of randomised trials. BMJ 2008;336:1121-3.

8. Tight blood pressure control and risk of macrovascular and microvascular complications in type 2 diabetes: UKPDS 38. UK Prospective Diabetes Study Group. BMJ 1998;317:703-13.

9. Hansson L, Zanchetti A, Carruthers SG, et al. Effects of intensive blood-pressure lowering and low-dose aspirin in patients with hypertension: principal results of the Hypertension Optimal Treatment (HOT) randomised trial. HOT Study Group. Lancet 1998;351:1755-62.

10. McAlister FA, Robitaille C, Gillespie C, et al. The impact of cardiovascular risk-factor profiles on blood pressure control rates in adults from Canada and the United States. Can J Cardiol 2013;29:598-605.

11. Sarafidis PA, Bakris GL. State of hypertension management in the United States: confluence of risk factors and the prevalence of resistant hypertension. J Clin Hypertens (Greenwich) 2008;10:130-9.

12. Persell SD. Prevalence of resistant hypertension in the United States, 2003-2008. Hypertension 2011;57:1076-80.

13. Egan BM, Zhao Y, Axon RN, Brzezinski WA, Ferdinand KC. Uncontrolled and apparent treatment resistant hypertension in the United States, 1988 to 2008. Circulation 2011;124:1046-58.

14. Roger VL, Go AS, Lloyd-Jones DM, et al. Heart disease and stroke statistics—2012 update: a report from the American Heart Association. Circulation 2012;125:e2-220.

15. Heidenreich PA, Trogdon JG, Khavjou OA, et al. Forecasting the future of cardiovascular disease in the United States: a policy statement from the American Heart Association. Circulation 2011;123:933-44.

16. Wagner EH, Austin BT, Von Korff M. Organizing care for patients with chronic illness. Milbank Q 1996;74:511-44.

17. Wagner EH, Warner JT, Slome C. Medical care use and hypertension. Med Care 1980;18:1241-50.

18. Kaczorowski J, Del Grande C, Nadeau-Grenier V. Community-based programs to improve prevention and management of hypertension: recent Canadian experiences, challenges, and opportunities. Can J Cardiol 2013;29:571-8.

19. Tobe SW, Moy Lum-Kwong M, Von Sychowski S, Kandukur K. Hypertension management initiative: qualitative results from implementing clinical practice guidelines in primary care through a facilitated practice program. Can J Cardiol 2013;29:632-5.

20. Nelson SA, Dresser GK, Vandervoort MK, et al. Barriers to blood pressure control: a STITCH substudy. J Clin Hypertens (Greenwich) 2011;13:73-80.

21. Alexander M, Gordon NP, Davis CC, Chen RS. Patient knowledge and awareness of hypertension is suboptimal: results from a large health maintenance organization. J Clin Hypertens (Greenwich) 2003;5:254-60.

22. Egan BM, Lackland DT, Cutler NE. Awareness, knowledge, and attitudes of older Americans about high blood pressure: implications for health care policy, education, and research. Arch Int Med 2003;163:681-7.

23. Williams MV, Baker DW, Parker RM, Nurss JR. Relationship of functional health literacy to patients' knowledge of their chronic disease. A study of patients with hypertension and diabetes. Arch Int Med 1998;158:166-72.

24. Jaffe MG, Lee GA, Young JD, Sidney S, Go AS. Improved blood pressure control associated with a large-scale hypertension program. JAMA 2013;310:699-705.

25. Yeh RW, Sidney S, Chandra M, et al. Population trends in the incidence and outcomes of acute myocardial infarction. N Engl J Med 2010;362:2155-65.

26. Koebnick C, Langer-Gould AM, Gould MK, et al. Sociodemographic characteristics of members of a large, integrated health care system: comparison with US Census Bureau data. Perm J 2012;16:37-41.

27. Sim JJ, Bhandari SK, Shi J, et al. Characteristics of resistant hypertension in a large, ethnically diverse hypertension population of an integrated health system. Mayo Clin Proc 2013;88:1099-107.

28. Kanter MH, Lindsay G, Bellows J, Chase A. Complete Care at Kaiser Permanente: transforming chronic and preventive care. Jt Comm J Qual Patient Saf 2013;39:484-94.

29. Selby JV, Lee J, Swain BE, et al. Trends in time to confirmation and recognition of new-onset hypertension, 2002-2006. Hypertension 2010;56:605-11.

30. Chapman JA, Johnson JA. On the spot! Peer validation of BP measurement competence. Nurs Manage 2013;44:22-4.

31. Ringerman E, Flint LL, Hughes DE. An innovative education program: the peer competency validator model. J Nurses Staff Dev 2006;22:114-21 [quiz 22-3].

32. Handler J, Zhao Y, Egan BM. Impact of the number of blood pressure measurements on blood pressure classification in US adults: NHANES 1999-2008. J Clin Hypertens (Greenwich) 2012;14:751-9.

33. Feldman RD, Zou GY, Vandervoort MK, et al. A simplified approach to the treatment of uncomplicated hypertension: a cluster randomized, controlled trial. Hypertension 2009;53:646-53.

34. Byrd JB, Zeng C, Tavel HM, et al. Combination therapy as initial treatment for newly diagnosed hypertension. Am Heart J 2011;162:340-6.

35. Cushman WC, Ford CE, Cutler JA, et al. Success and predictors of blood pressure control in diverse North American settings: the antihypertensive and lipid-lowering treatment to prevent heart attack trial (ALLHAT). J Clin Hypertens (Greenwich) 2002;4:393-404.

36. Sim JJ, Bhandari SK, Shi J, et al. Plasma renin activity (PRA) levels and antihypertensive drug use in a large healthcare system. Am J Hypertens 2011;25:379-88.

37. Fung V, Huang J, Brand R, Newhouse JP, Hsu J. Hypertension treatment in a medicare population: adherence and systolic blood pressure control. Clin Ther 2007;29:972-84.

38. Harrison TN, Ho TS, Handler J, et al. A randomized controlled trial of an automated telephone intervention to improve blood pressure control. J Clin Hypertens (Greenwich) 2013;15:650-4.

39. Kanter M, Martinez O, Lindsay G, Andrews K, Denver C. Proactive office encounter: a systematic approach to preventive and chronic care at every patient encounter. Perm J 2010;14:38-43.

40. Yiannakopoulou E, Papadopulos JS, Cokkinos DV, Mountokalakis TD. Adherence to antihypertensive treatment: a critical factor for blood pressure control. Eur J Cardiovasc Prev Rehabil 2005;12:243-9.

41. Bramley TJ, Gerbino PP, Nightengale BS, Frech-Tamas F. Relationship of blood pressure control to adherence with antihypertensive monotherapy in 13 managed care organizations. J Manag Care Pharm 2006;12:239-45.

42. Garg JP, Elliott WJ, Folker A, Izhar M, Black HR. Resistant hypertension revisited: a comparison of two university-based cohorts. Am J Hypertens 2005;18:619-26.

43. Ho PM, Bryson CL, Rumsfeld JS. Medication adherence: its importance in cardiovascular outcomes. Circulation 2009;119:3028-35.

44. Ho PM, Rumsfeld JS, Masoudi FA, et al. Effect of medication nonadherence on hospitalization and mortality among patients with diabetes mellitus. Arch Int Med 2006;166:1836-41.

45. Van Wijk BL, Klungel OH, Heerdink ER, de Boer A. Rate and determinants of 10-year persistence with antihypertensive drugs. J Hypertens 2005;23:2101-7.

46. Mazzaglia G, Mantovani LG, Sturkenboom MC, et al. Patterns of persistence with antihypertensive medications in newly diagnosed hypertensive patients in Italy: a retrospective cohort study in primary care. J Hypertens 2005;23:2093-100.

47. Vrijens B, Vincze G, Kristanto P, Urquhart J, Burnier M. Adherence to prescribed antihypertensive drug treatments: longitudinal study of electronically compiled dosing histories. BMJ 2008;336:1114-7.

48. Bosworth HB, Olsen MK, Goldstein MK, et al. The veterans' study to improve the control of hypertension (V-STITCH): design and methodology. Contemp Clin Trials 2005;26:155-68.

49. Roumie CL, Elasy TA, Greevy R, et al. Improving blood pressure control through provider education, provider alerts, and patient education: a cluster randomized trial. Ann Int Med 2006;145:165-75.

50. Kovesdy CP, Bleyer AJ, Molnar MZ, et al. Blood pressure and mortality in U.S. Veterans with chronic kidney disease: a cohort study. Ann Int Med 2013;159:233-42.

51. Alderman MH, Cohen HW, Sealey JE, Laragh JH. Plasma renin activity levels in hypertensive persons: their wide range and lack of suppression in diabetic and in most elderly patients. Am J Hypertens 2004;17:1-7.

52. Handler J, Lackland DT. Translation of hypertension treatment guidelines into practice: a review of implementation. J Am Soc Hypertens 2011;5:197-207.

53. Fletcher RD, Amdur RL, Kolodner R, et al. Blood pressure control among US veterans: a large multiyear analysis of blood pressure data from the Veterans Administration health data repository. Circulation 2012;125:2462-8.

54. U.S. Department of Veterans Affairs. Quality of Care. Available at: http://www.va.gov/QUALITYOFCARE/initiatives/compare/high-blood-pressure-control.asp. Accessed December 3, 2013.

55. Bex SD, Boldt AS, Needham SB, et al. Effectiveness of a hypertension care management program provided by clinical pharmacists for veterans. Pharmacotherapy 2011;31:31-8.

56. Zhou YY, Kanter MH, Wang JJ, Garrido T. Improved quality at Kaiser Permanente through e-mail between physicians and patients. Health Aff (Millwood) 2010;29:1370-5.